Monomials
on one page

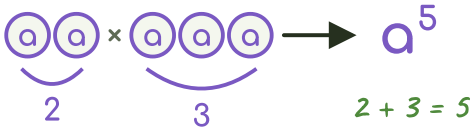


Hi! Bulya, Prickles, Muddle and I have drawn everything important about monomials. The arrows show you what goes where.

Powers: a quick reminder 1

A power is a short way to write repeated multiplication: $a^3 = a \times a \times a$. When you multiply powers, **add the indices**.

$a^2 \times a^3 =$

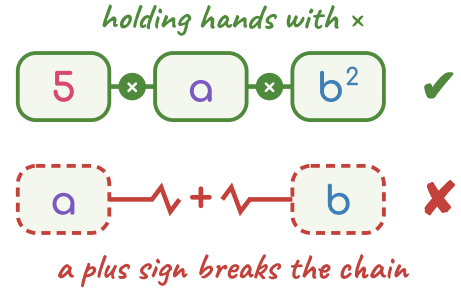


$(a^2)^3 = a^2 \times a^2 \times a^2 = a^6$

multiply the powers: 2 × 3 = 6

What is a monomial? 2

A **monomial** is a single term: numbers, letters and their powers multiplied together. Only ×, no + or −.



These are monomials too: **7**, **x**, **y³**, **−ab**, even **0**. A minus in front belongs to the number, so it is fine.



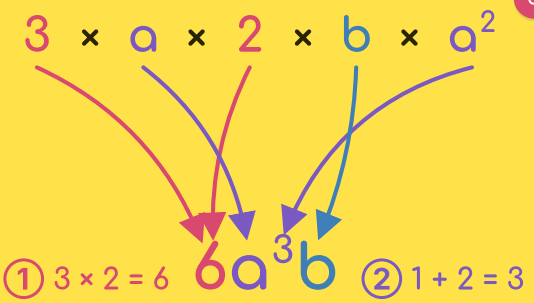
A monomial is like me and Ponya holding paws. Put a plus between us and we are two separate guinea pigs!

The big rule: simplest form 3

① **Multiply all the numbers and put the answer first.**
② **Collect each letter once: add its indices.**

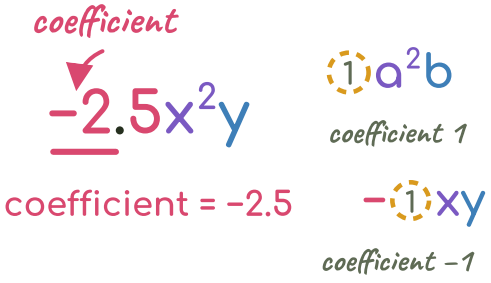


Number at the front, each letter only once. Just like my tidy cupboard!



The coefficient 4

The number at the front of a monomial in simplest form. If you cannot see a number, a 1 is hiding there.

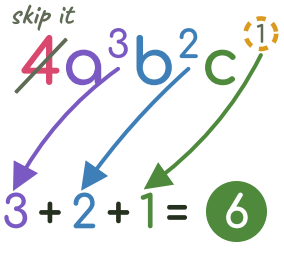




An invisible 1 always sits in front of the letters. Sometimes with a minus!

The degree of a monomial 5

Add up the indices of **all the letters**. Ignore the number at the front.



c means c^1. A number on its own (like 7) has degree 0.

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Multiplying monomials

Multiply the numbers together. Combine matching letters: **add the indices**.

Check: the degree of the answer equals the degrees added. 3 + 4 = 7, and $12a^3b^4$ has degree 7 too.

$$(3a^2b)\times(4ab^3)$$

3×4 2+1 1+3

$$12a^3b^4$$

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A monomial to a power

The index outside the bracket goes to **every** factor inside. Multiply the letters' indices.



The index outside the bracket is like an umbrella: it covers everyone inside.

A minus disappears with an even power: $(-2)^2 = 4$. It stays with an odd power: $(-2)^3 = -8$.

umbrella

$$(-2a^3b)^2$$

$(-2)^2=4$ 3×2 1×2

$$4a^6b^2$$

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Working backwards

Finding a missing factor? **Divide the numbers, subtract the indices**. Finding what was squared? **Halve the indices**.

$$3a^2b\times A = 12a^5b^3$$

12÷3 5−2 3−1

$$A = 4a^3b^2$$

$$49a^6b^4 = (?)^2$$

7×7=49 6÷2 4÷2

$$(7a^3b^2)^2$$

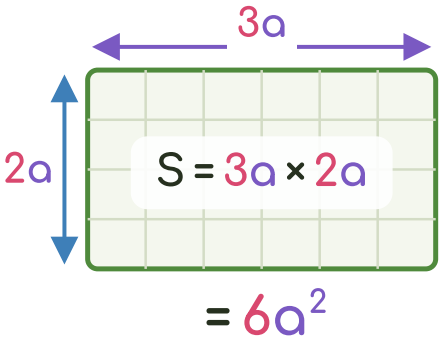
For a cube, divide the indices by 3 and find the number that, times itself three times, gives the one you need.

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Monomials in real life

A new run for Ponya and Bulya: 3a long and 2a wide. Its area is a monomial!

If a = 1 m, the area is 6 m². If a = 2 m, it is 6 × 4 = 24 m².



$$= 6a^2$$



Check yourself: one number at the front, each letter just once. Then it is all tidy and Prickles is happy!

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Built on his slides and lessons.